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A Prompt Design

Tables 6 and 7 present the prompts used in the baseline, including Single Agent, Single Agent with Multiple Role Prompt, and LLM Discussion. In these tables, the sections highlighted in yellow indicate the task descriptions. The current example shows the AUT task, but the highlighted part will change depending on the specific task. Descriptions for different tasks are provided in Table 8. Additionally, while the original dataset from LLM Discussion (Lu et al., 2024) originally asked models to generate “as many as possible" responses, we observed that this instruction posed challenges for smaller language models. To ensure fairness and comparability, we modified these prompts to instead request five responses. In the LLM Discussion setting, prompts vary not only with the task but also with the role assignments of different agents. These role-specific variations are highlighted in blue, and the full list of role settings is provided in Table 9. The prompt templates used for LLM evaluation are presented in Tables 10, 11, and 12.

B Methodology for Persona Vector Extraction

To extract our persona vectors, we adapt the contrastive activation methodology from Chen et al. (2025), focusing on the complex professional roles defined in Lu et al. (2024). Unlike the original method which targeted simple personality traits like evil, humorous, our work required vectors for roles like creative professional or environmentalist to align with our creativity benchmarks. This necessitated the creation of a bespoke dataset and a complete pipeline for extraction, which we detail below.

The process begins with generating two contrasting sets of model responses for each target persona P. First, we used Claude-sonnet-4 to create five distinct positive system prompts that exemplify each persona’s unique traits (e.g., an environmentalist’s perspective on a problem). These prompts are detailed in Table 13 and 14.

Next, we generated responses to 20 trait-eliciting questions using both the positive prompts and a standard neutral prompt (e.g., "You are a helpful assistant."). To ensure a clear distinction between persona-aligned and neutral behavior, each response was then scored for trait expression (0-100) by an LLM-judge (GPT-4o-mini), a technique validated by Chen et al. (2025). Based on these scores, we curated two final datasets:

- The **positive-expression set** (D_P^+), containing positive-prompt responses with scores greater than 50.
- The **neutral-expression set** (D_P^-), containing neutral-prompt responses with scores less than 50.

Finally, we calculate the persona vector by contrasting the internal activations of these two sets. For each response $\mathbf{x}$ in these datasets, we extract its token-averaged residual stream activation at a specific layer l, denoted as $\vec{a}^{(l)}(\mathbf{x})$. The persona vector $\vec{v}_P^{(l)}$ is then defined as the difference between the mean activation vectors of the two sets:

$$\vec{v}_P^{(l)} = \frac{1}{|D_P^+|} \sum_{\mathbf{x} \in D_P^+} \vec{a}^{(l)}(\mathbf{x}) - \frac{1}{|D_P^-|} \sum_{\mathbf{x} \in D_P^-} \vec{a}^{(l)}(\mathbf{x}) \tag{6}$$

This procedure is repeated for all layers, generating a unique candidate vector for each layer, from which an optimal layer is later selected for our experiments.

C Steering Layer and Coefficient Choice

Our experiments adopted default settings of persona vectors steering as Chen et al. (2025), where layer=20, and α=2.0. We further conduct a more comprehensive study across **4 layers (12-24)** and **7 α values (0.1–5.0)** on all tested models, which are shown in Table 15 and Table 16. Below are the observations:

- Sensitivity to layer: Across all models, Layer 20 consistently yields near-optimal performance, confirming it as a robust cross-model choice.
- Sensitivity to α: Models exhibit high stability within the range of $1.0 \leq \alpha \leq 2.5$. Our chosen $\alpha = 2.0$ falls within this optimal window, whereas extreme values ($\alpha = 0.1$ or $\alpha = 5$) lead to performance degradation.

D Linguistic Diversity $\neq$ True Creativity

To determine whether the LLM judge merely rewards lexical variety rather than genuine creative merit, we compared BILLY against a high-temperature baseline (SA, $T = 1.0$). While increasing generation temperature maximizes linguistic diversity, our analysis reveals that such stochastic variety does not yield a proportional increase in creativity scores. This finding aligns with observations from Lu et al. (2024), which noted that high-temperature outputs in commercial models often lead to noisy diversity rather than enhanced quality.

As summarized in Table 2, BILLY consistently outperforms the high-diversity SA ($T = 1.0$) baseline across all models. For instance, on the Qwen-2.5-7B (AUT Originality) task, BILLY achieves a score of 4.71±0.45, significantly higher than the baseline's 4.15±0.51. These results suggest that the LLM judge possesses the ability to distinguish between mere linguistic diversity and novelty indicative of true creativity. Furthermore, our Human Evaluation results in Table 3 support these findings, showing that BILLY effectively outperforms baselines in both Originality and Elaboration.

E Amortized Cost of Persona Vector Generation

An advantage of BILLY is that the initial, one-time cost of generating the persona vectors is amortized over subsequent uses. As shown in Figure 5, the average cost per query for BILLY decreases dramatically as the number of queries increases. All inference experiments were conducted on a machine with an NVIDIA RTX 6000 Ada GPU running Ubuntu 22.04.

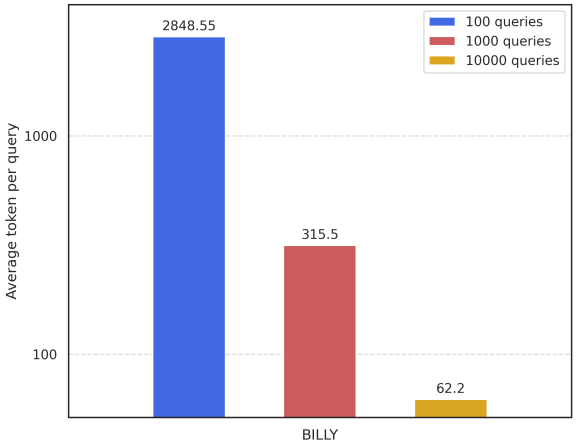

Figure 5: **Amortized Average Input Token Per Query.** Token cost per query would be amortized with more inference frequency.

E.1 Comparative Token Cost at Scale

Figure 6 provide a direct comparison of the average token costs per query for each method, calculated at a scale of 10,000 queries. The data clearly illustrates the efficiency improvement of BILLY over the LLM Discussion baseline.

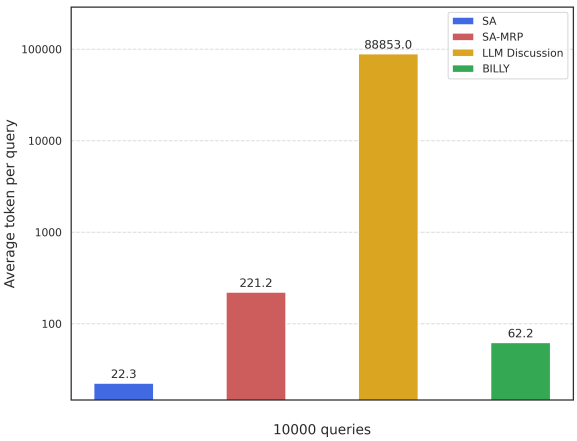

Figure 6: **Average Input Token Count.** Input token amount of BILLY per query is close to single agent at the scale of 10000 queries.

F Persona Vector Combinations for Discussion Section

Section 5.2 (Table 2) systematically varies the number of merged vectors. The specific combinations

of personas used for each experimental condition are detailed in Table 17.

G Comprehensive Qualitative Result

We present the qualitative results for same question, *"Healthcare Innovation (Reimagine the Hospital) Redesign the future hospital experience for patients, families, and staff. Address flows, safety, data, well-being, equity, and feasibility."*, of the model steered by single CRE vector, single ENV vector, merged CRE and ENV vector, and prompt the model to be CRE and ENV in Table 18, 19, 20, and 21. Qualitative analysis of the outputs (Tables 18, 19, and 20) reveals a functional distinction between the personas. The creative professional persona primarily influences the linguistic style of the response, introducing more artistic and descriptive language. In contrast, the environmentalist persona affects the core content, shaping the ideas and structure of the output. Consequently, the merged vector, BILLY (CRE + ENV), synthesizes these roles: it generates responses with the thematic structure of the environmentalist, but in the polished and creative style of the creative professional.

H Trait Expression Scores Results

Beyond the sheer quantity of vectors, another significant findings emerge from analyzing how different methods affect the persona of output responses. Our experiments, conduct on a trait-neutral dataset (Table 23) and evaluate using a distributed scoring method, using Gemini-2.5-pro, Gemini-2.0-flash, and GPT-4o-mini as LLM judges and average the given scores to reduce the variance and bias. Specifically, we ask LLM judges to distribute 100 scores to each persona with evaluation prompts in Table 24. Table 22 reveals that SA (Baseline) and SA-MRP (ENV + CRE + FUT) have almost the same trait expression score among all of the personas, which confirms that the ability of steering persona via prompt is limited. In contrast, while the model is steered by single vector (either CRE or ENV), the desired trait expression score is significantly higher than SA (Baseline). However, we find that BILLY operates not as a simple summation of traits, where vectors exhibit complex effects and collaboration as we discussed in Appendix G. For instance, while the BILLY (CRE + ENV) maintains the skeleton of environmentalist, it uses the creative term to decorate the responses, which we hypothesize would be more easily to be judge. As a result, the LLM judges give higher scores on creativity rather than environmentalist.

I Process Loss Analysis of LLM Discussion

The multi-agent discussion log (Table 25) serves as a compelling illustration of the spectrum of *process loss* within collaborative LLM systems. The example of "a book and a magazine" epitomizes severe process loss, particularly in the forms of coordination and motivational failures. Despite their distinct roles, the agents exhibit substantial redundancy, repeatedly converging on overlapping concepts such as narrative and visual appeal across multiple rounds. More critically, they fail to achieve synthesis: each agent largely disregards the contributions of its peers, resulting in conversational stagnation where the final output is little more than a rephrasing of the initial proposal. Consequently, the collective performance is substantially less than the sum of its potential parts, reflecting agents that operate in parallel silos rather than as a cohesive team. This leads to redundancy, inflated time costs, and overall inefficiency.

J Chat Log of LLM Discussion

Table 26 shows the chat log of one agent in LLM Discussion. The complete agent discussion log for the Similarities Creativity task is provided in the supplementary materials due to its extensive nature. The file discussion_log.zip contains llm_discussion_chatlog.json, which documents the multi-agent interaction within the LLM Discussion framework.